Russell W. Mercer

Provo, UT · rmercer2@byu.edu · linkedin.com/in/russell-mercer · orcid.org/0009-0003-6387-969X · github.com/mercer-science

EDUCATION

Brigham Young University, Provo, UT January 2024 – April 2028 (Expected)

B.S. in Biochemistry, Minor in Computer Science (machine learning emphasis)

- Cumulative GPA 3.99; Science GPA 3.99
- Advanced biochemistry, pre-medical, and machine learning coursework

Medical College Admission Test (MCAT): 519, 96th percentile August 2026

RESEARCH EXPERIENCE

Biochemistry Research Assistant, Jensen Lab, BYU August 2026 – Present

Department of Chemistry and Biochemistry. Mentor: Grant J. Jensen, PhD, Dean, College of Physical and Mathematical Sciences; with Chitral Chatterjee

- Scoped and proposed two cryo-electron tomography projects, approved by the PI and awaiting institutional biosafety review: the surface architecture of *Bartonella henselae*, and the apical complex of *Babesia duncani*
- Testing whether a general-purpose AI coding agent can shorten the cryo-electron tomography learning curve for a new researcher without replacing the understanding the work normally builds
- Processing 39 tilt series of cryo-FIB-milled, exponentially growing *E. coli* through motion correction, alignment, reconstruction, missing-wedge correction, template matching, classification, and subtomogram averaging, ending in a ribosome neighbor analysis
- Recording which agent recommendations are accepted, overridden, or caught as errors, and packaging the workflow as a reusable agent skill

Clinical Research Assistant, UT Southwestern Medical Center May 2026 – Present

Radiation Oncology, Kenneth Westover, MD, PhD (May to August 2026); Medical Physics, Jie Deng, PhD (August 2026 to present)

- Started through the BYU Summer Premedical Research Internship, funded by the BYU Simmons Center for Cancer Research; on site in Dallas, TX through August 2026 and remote since
- Four studies: lung and pancreas diffusion-weighted MRI on the Elekta Unity MR-Linac, biology-guided radiotherapy on RefleXion, and radiolucent patient immobilization. First author on all four manuscripts
- Rebuilt the department's diffusion MRI processing pipeline into a web application with a bulk runner, viewer, threshold tool, and fit checker; ran 76 patients in about 2.5 days of server time, and hospital IT moved it to the Radiation Oncology Application Team for clinical use
- Wrote the C#, Python, and R extraction and figure pipelines behind the studies; abstracted clinical data in Epic; wrote the hand-off document and trained the postdoc who inherited the pipeline
- Offered a continuing position in Medical Physics after the summer internship

Chemistry Research Assistant, Stowers Lab, BYU April 2024 – May 2026

Department of Chemistry and Biochemistry. Mentor: Kara J. Stowers, PhD

- Synthesized surface-bound metal-organic framework thin films (CuBDC/DABCO, HKUST-1) by spin coating and studied their structure and decomposition
- Helped build, repair, and maintain a flow reactor and an ultra-high-vacuum chamber reaching about 10^{-10} Torr, with X-ray photoelectron spectroscopy and mass spectrometry
- Trained newer lab members in spin coating and SEM imaging
- Supported by a BYU Undergraduate Research Award; still contributing to a manuscript in preparation

PUBLICATIONS

Peer-Reviewed

Ralph AD, **Mercer RW**, Hall SB, Butikofer BC, Collins CS, Sturgill CJ, Minson PS, Stefik M, Stowers KJ. Ambient pressure fabrication of $Cu_xZn_{1-x}(BDC)_2DABCO$ thin films through spin-coating and their temperature-dependent structural development. *Langmuir*. 2026;42(6):4475–4488. doi:10.1021/acs.langmuir.5c05182

Submitted

Mercer RW, et al. Lung PULSAR diffusion-weighted MRI on the Elekta Unity MR-Linac. UT Southwestern. Submitted to the *International Journal of Radiation Oncology, Biology, Physics*, October 2026.

In Co-Author Circulation

Mercer RW, et al. Biology-guided radiotherapy on RefleXion: dosimetry and outcomes. UT Southwestern.

Mercer RW, et al. Radiolucent patient immobilization: a dosimetric comparison. UT Southwestern.

In Preparation

Mercer RW, et al. Pancreas diffusion-weighted MRI on the Elekta Unity MR-Linac. UT Southwestern.

Mercer RW, Chatterjee C, Jensen GJ. Using an AI coding agent to learn cryo-electron tomography processing. Jensen Lab, BYU.

Co-author, surface chemistry in the Stowers Lab ultra-high-vacuum chamber. BYU.

Co-author on two further UT Southwestern studies.

Author order on manuscripts still in review is not final.

PRESENTATIONS AND ABSTRACTS

Mercer RW, Chatterjee C, Jensen GJ. Can an AI agent collapse the cryo-ET learning curve? Poster, Utah CryoEM Symposium, Brigham Young University, Provo, UT. — October 2026

Invited guest talk on UT Southwestern research and AI in research. HL 385R, Healthcare Research Hub, BYU Healthcare Leadership, Marriott School of Business, Provo, UT. — October 2026

UT Southwestern radiation oncology research. Poster, BYU Medical Society, Provo, UT. — August 2026

UT Southwestern radiation oncology research. Oral presentation, BYU Simmons Center for Cancer Research, Provo, UT. — August 2026

Metal-organic framework thin films. American Chemical Society Spring Meeting, Atlanta, GA. — March 2026

Metal-organic framework thin films. BYU Student Research Conference, Provo, UT. — February 2026

Fast room temperature synthesis and characterization of a surface bound metal organic framework without the use of a self-assembling monolayer. BYU Student Research Conference, Provo, UT. — March 2025

HONORS, AWARDS, AND FUNDING

SCCR Fellow, BYU Simmons Center for Cancer Research (funded summer research at UT Southwestern) — 2026

Outstanding Analytical Chemist Award, BYU — 2025

Dean's List, BYU (Winter, Spring, and Fall 2024; Fall 2025) — 2024 – 2025

Undergraduate Research Award, BYU (awarded each semester) — 2024 – 2026

Stoney Black Flight of the Phoenix Award, Air Academy High School Marching Band — 2021

Eagle Scout, Boy Scouts of America — 2020

CLINICAL EXPERIENCE

Medical Assistant, Health Plus Clinic, Springville, UT — July 2025 – April 2026

- Assisted a D.O. and a P.A. with patient care and clinical procedures
- Kept medical records accurate and compliant; completed HIPAA and medical assistant training

Volunteer, Utah State Hospital, Provo, UT May 2025 – Present

- Physical Therapy (May 2025 to present): guide 10+ mental health patients through therapeutic exercise and help the physical therapist keep sessions safe and encouraging
- Pediatric Center (September 2025 to January 2026): led group activities for pediatric mental health patients
- Pediatric Mentor (December 2025 to May 2026): met one-on-one with pediatric mental health patients

PHYSICIAN SHADOWING

Steven M. Grover, Robotic Gynecologic Surgery, Utah Valley Hospital, 3.5 hours — October 2026

Thomas Fowler, Allergy, 4 hours — June 2026

Stanford C. Taylor, Retina Surgery, Pueblo, CO, 6 hours — April 2026

Matheson Harris, Plastic Surgery (Facial Reconstruction), Utah Surgical, 7.5 hours — November 2025

Michael Benson, Psychiatry, Utah State Hospital, 8 hours — August 2025

Dan Baer, Pain Management (PM&R), 3.5 hours — July 2025

James Barrott, General Surgery; Joshua Jaramillo, Trauma Surgery and ICU; Eric Cannon, Anesthesia. Scheduled. — November – December 2026

LEADERSHIP AND TEACHING

Officer, BYU AI Association September 2026 – Present

- Lead the weekly AI in Research meeting
- Speak about AI in research to other colleges across the university, and meet with professors and labs about using it in their own work
- Represented the association at BYU's AI for Social Innovation Conference, on the Data Sovereignty and Inclusion working table

Pre-Med Peer Mentor, BYU Pre-Med Association October 2026 – Present

- Mentor pre-med students on coursework, research, and clinical experience

Lead Program Director, BYU Y-Serve, Early Learning Essentials August 2024 – Present

- Program Director from August 2024; Lead Program Director since March 2025
- Recruit volunteers, organize events, and manage communication for a program serving preschoolers from low-income families

University Peer Mentor, BYU Office of First Year Experience June 2024 – February 2025

- Co-taught two sections of Introduction to University (UNIV 101) with faculty
- Mentored 45 first-year students through regular one-on-one meetings; completed QPR gatekeeper training

Full-Time Missionary, California Riverside Mission October 2021 – December 2023

The Church of Jesus Christ of Latter-day Saints

- Led and supported small groups of fellow volunteers
- Planned and gave 80+ trainings on teaching skills to groups of 8 to 100 volunteers; mentored four new volunteers one-on-one
- Worked in the mission office coordinating housing and travel logistics and managing technology

Squad Leader, Air Academy High School Marching Band, Colorado Springs, CO May 2020 – May 2021

- Led the trumpet section; the band won two state championships and marched in the 2018 Rose Parade

SERVICE

Volunteer, Global Early Detection Initiative, Utah Chapter (community outreach) — August 2026 – Present

Church Instructor, The Church of Jesus Christ of Latter-day Saints — August 2026 – Present

Congregation Activities Committee and Leadership Council — September 2025 – April 2026

Congregation Clerk (records) and Leadership Council — May 2024 – August 2024

Boy Scouts of America; Eagle Scout project for the National Mill Dog Rescue — 2014 – 2020

CERTIFICATIONS AND TRAINING

Thermo Fisher Scientific "Resolution Evolution" cryo-EM training (Vitrobot, Krios) — October 2026

Epic for Research Coordinators, UT Southwestern — May 2026

SCCR Research Ethics Seminar, BYU — April 2026

NHCO Certified Phlebotomy Technician (two-year certification, expired) — June 2024 – June 2026

QPR Suicide Prevention Gatekeeper Training, BYU — 2024

CONFERENCES ATTENDED

AI for Social Innovation Conference, BYU Ballard Center, Aspen Grove, UT — October 2026

Utah CryoEM Symposium, Brigham Young University — October 2026

American Chemical Society Spring Meeting, Atlanta, GA — March 2026

SKILLS

Laboratory: bacterial culture and plasmid work, SEM, ultra-high-vacuum systems, XPS, mass spectrometry, spin coating, flow reactor operation, Vitrobot, Krios cryo-TEM

Computational: Python, R, C#, Java, shell, Git and GitHub, ImageJ, Epic, cryo-ET processing software, DICOM and medical imaging pipelines, GPU workflows; AI coding agents since 2024, including agent skills and hooks

Communication: first author on four manuscripts and a co-author on several more in preparation; six research presentations and an invited guest talk; figure, poster, and interface design

ACTIVITIES

BYU Fencing Club — August 2026 – Present

Intramural Ultimate Frisbee, BYU — May 2025 – December 2025